

Set A

Question - 1

(a)

$$\Delta_{ADC} = \frac{V_{max} - V_{min}}{2^N}$$

$$= \frac{12 - 0}{2^2}$$

$$\Delta_{ADC} = 3V$$

when,

$$V_{in} = 0V$$

(reaches lower peak)

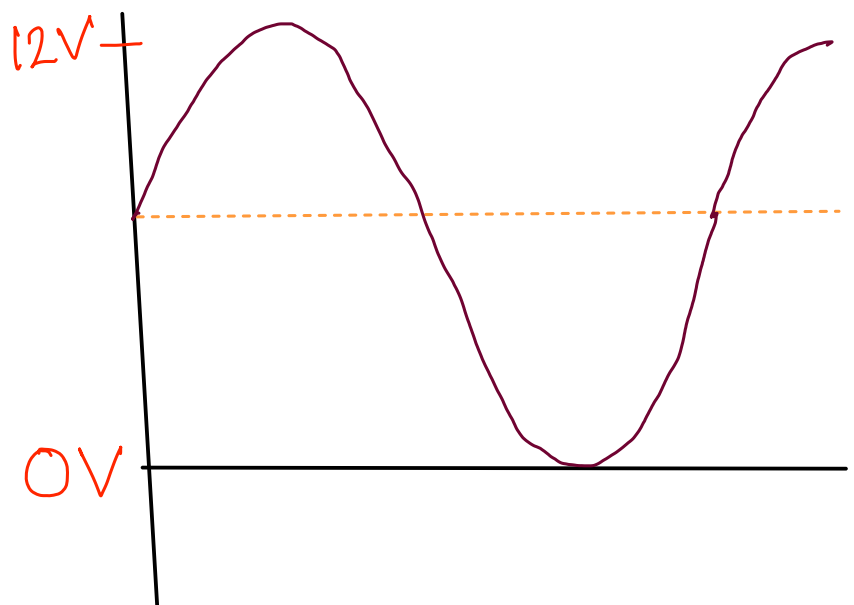
$$Y_1 = 0 ; Y_0 = 0$$

when,

$$V_{in} = 12V$$

(reaches higher peak)

$$Y_1 = 1 ; Y_0 = 1$$



Q. range	Q. level	encoded binary
0-3	1.5	00
3-6	4.5	01
6-9	7.5	10
9-12	10.5	11

Now,

$$V_{out} = -\frac{R_F \cdot V_{ref}}{R} \left(Y_1 + \frac{Y_0}{2} \right)$$

$$\therefore V_{00} = \frac{-5 \times (3V)}{5} \left(0 + \frac{0}{2} \right)$$
$$= 0V$$

$$\therefore V_{II} = \frac{-5 \times (3V)}{5} \left(1 + \frac{1}{2}\right)$$

$$= -4.5V$$

when $V_{in} = 0$	when $V_{in} = 12V$
$\% \text{ error} = \frac{0 - 0 }{0} \times 100\%$ $= \text{undefined}$	$\% \text{ error} = \frac{12 - -4.5 }{12} \times 100\%$ $= 62.5\%$

(b)

ADC step size	DAC step size
$\Delta_{ADC} = \frac{12 - 0}{2^2}$ $= 3V$	$V_{OI} = -\frac{5 \times (3V)}{5} \left(0 + \frac{1}{2}\right)$ $= -1.5V$

Yes, they are different, Suggested steps to match them $\rightarrow$ we shall take magnitude value
 $|-3| = 3V$; $|-1.5| = 1.5V$

① V_{Ref} increase $\rightarrow$

$$V_{OI} = \left| -\frac{5 \times V_{Ref}}{5} \left(0 + \frac{1}{2}\right) \right|$$

$$\Rightarrow 3 = V_{Ref} \times \frac{1}{2}$$

$$\therefore V_{Ref} = 6V \left[V_i = V_o = 6V \text{ for logic high now} \right]$$

② R_F increase $\rightarrow$

$$V_{OI} = \left| -\frac{R_F' \times 3V}{5} \left(0 + \frac{1}{2}\right) \right|$$

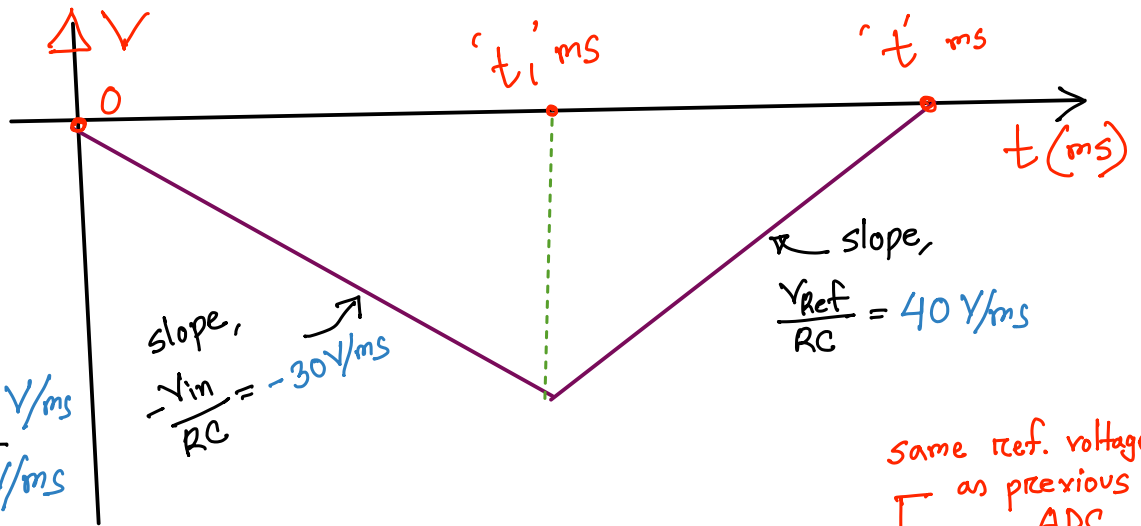
$$\Rightarrow 3 = R_F' \times \frac{3}{10}$$

$$\therefore R_F' = 10k\Omega$$

(c)

Thus,

$$\frac{-\frac{V_{in}}{RC}}{\frac{V_{ref}}{RC}} = \frac{-30 \text{ V/ms}}{40 \text{ V/ms}}$$



slope,
 $-\frac{V_{in}}{RC} = -30 \text{ V/ms}$

slope,
 $\frac{V_{ref}}{RC} = 40 \text{ V/ms}$

same ref. voltage
as previous
ADC

$$\Rightarrow \frac{V_{in}}{V_{ref}} = \frac{3}{4} \quad \Rightarrow V_{in} = \frac{3}{4} \times V_{ref} = \frac{3}{4} \times 12 \text{ V}$$

$\therefore V_{in} = 9 \text{ V}$ is the specific input,

we know, $V_{in} = m \times \text{step-size}$

$$\Rightarrow m = \frac{V_{in}}{\text{step-size}}$$

$$\Rightarrow m = \frac{9 \text{ V}}{\frac{12-0}{2^8}}$$

8-bit dual slope ADC

$$\therefore m = 192$$

$(192)_{10} = (11000000)_2$ is the corresponding digital o/p of the $V_{in} = 9 \text{ V}$

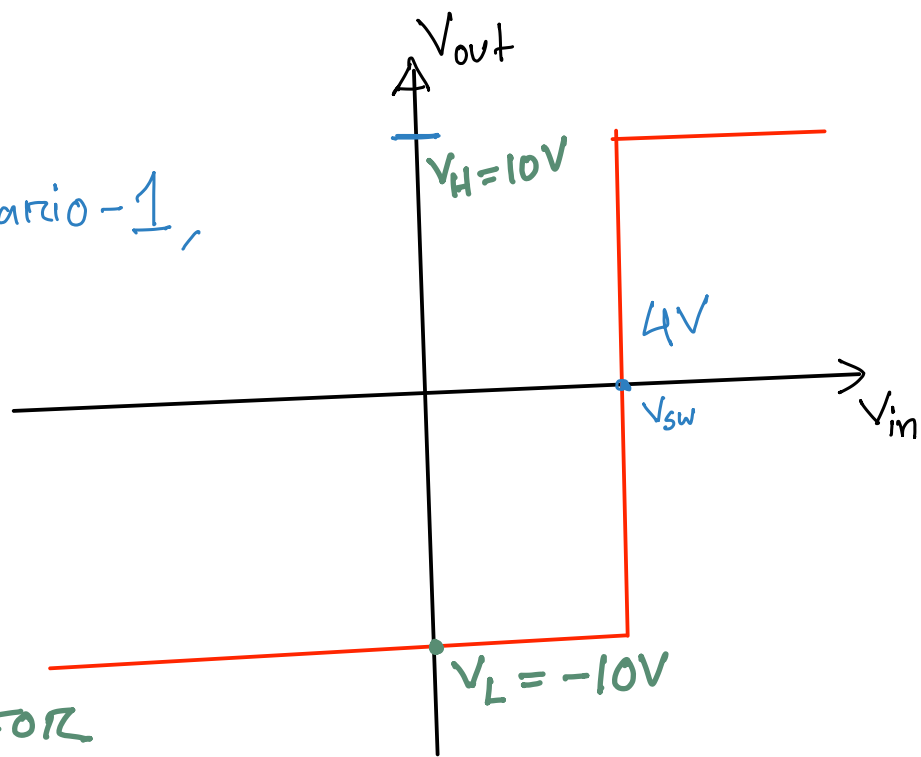
please
note

if your formula is, $V_{in} = (m+1) \times \text{step size}$
then, $m = 191$

$$\therefore m = 10111111$$

Question-2

(a) As per the scenario-1,



This is the VTC of a
Non-inverting comparator

$$\text{Thus, } -\frac{R_2}{R_1} \cdot V_{Ref} = V_{sw}$$

$$\Rightarrow -\frac{R_2}{R_1} (-5) = 4$$

$$\Rightarrow \frac{R_2}{R_1} = \frac{4}{5}$$

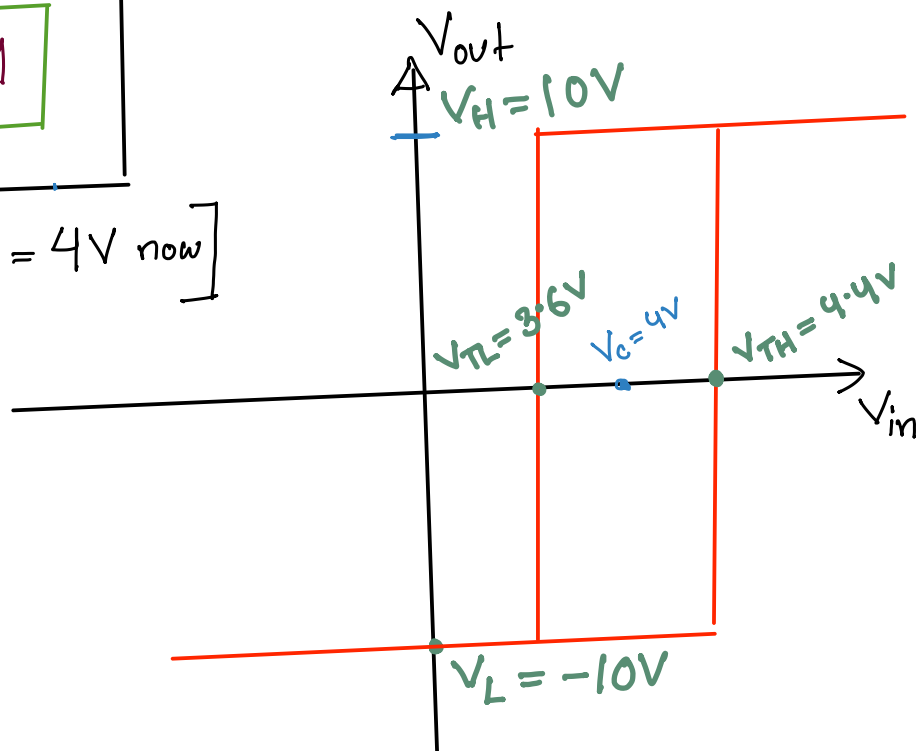
$$\Rightarrow \frac{R_1}{R_2} = \frac{5}{4}$$

$$\therefore R_1 : R_2 = 5 : 4$$

(b)

0.8V peak to peak noise
consideration = hysteresis of
0.8V

So, at both sides of $V_{sw} = 4V$,
there will be $\frac{0.8}{2} = 0.4$ volt of
margin



$$\therefore V_{TH} = 4 + 0.4$$

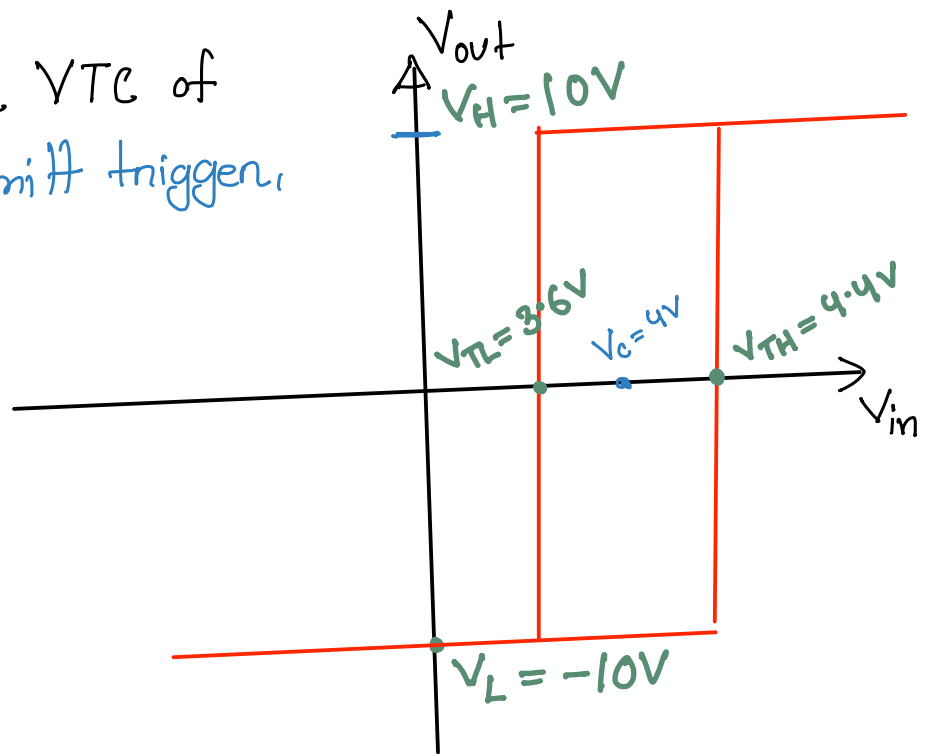
$$= \underline{4.4 \text{ volt}}$$

$$[V_c = 4V \text{ now}]$$

$$\therefore V_{TL} = 4 - 0.4$$

$$= \underline{3.6 \text{ volt}}$$

(c) This is now the VTC of a Non-inverting schmitt trigger, with Ref. voltage,



we know formulas,

$$V_{TH} = \frac{R_1 + R_2}{R_2} V_{Ref} - \frac{R_1}{R_2} V_L \quad \text{..... (i)}$$

$$V_{TL} = \frac{R_1 + R_2}{R_2} V_{Ref} + \frac{R_1}{R_2} V_H \quad \text{..... (ii)}$$

$$\begin{aligned} (V_{TH} - V_{TL}) &= -\frac{R_1}{R_2} V_L + \frac{R_1}{R_2} V_H \\ &= \frac{R_1}{R_2} (V_H - V_L) \end{aligned}$$

$$\Rightarrow \frac{R_1}{R_2} = \frac{V_{TH} - V_{TL}}{V_H - V_L}$$

$$\Rightarrow \frac{R_1}{R_2} = \frac{4.4 - 3.6}{10 - (-10)}$$

$$\Rightarrow \frac{R_1}{R_2} = \frac{0.8}{20}$$

$$\Rightarrow \frac{R_1}{R_2} = \frac{1}{25}$$

$$\therefore R_2 = 25 R_1$$

$$\text{let, } R_1 = 1k\Omega$$

$$\therefore R_2 = 25k\Omega$$

[(i) - (ii) is done]

stiff voltage

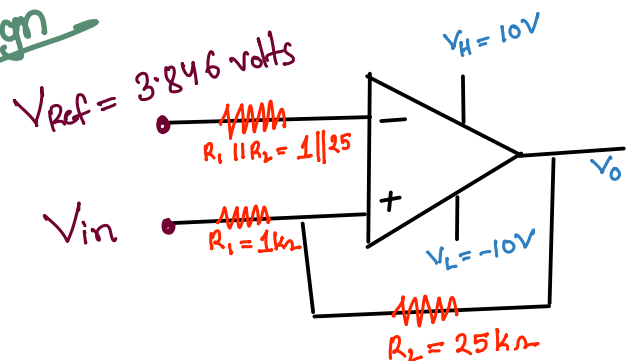
$$\text{Now, } \frac{R_1 + R_2}{R_2} V_{Ref} = \frac{4.4 + 3.6}{2}$$

$$\Rightarrow \frac{1 + 25}{25} \cdot V_{Ref} = 4$$

$$\Rightarrow V_{Ref} = \frac{4 \times 25}{26}$$

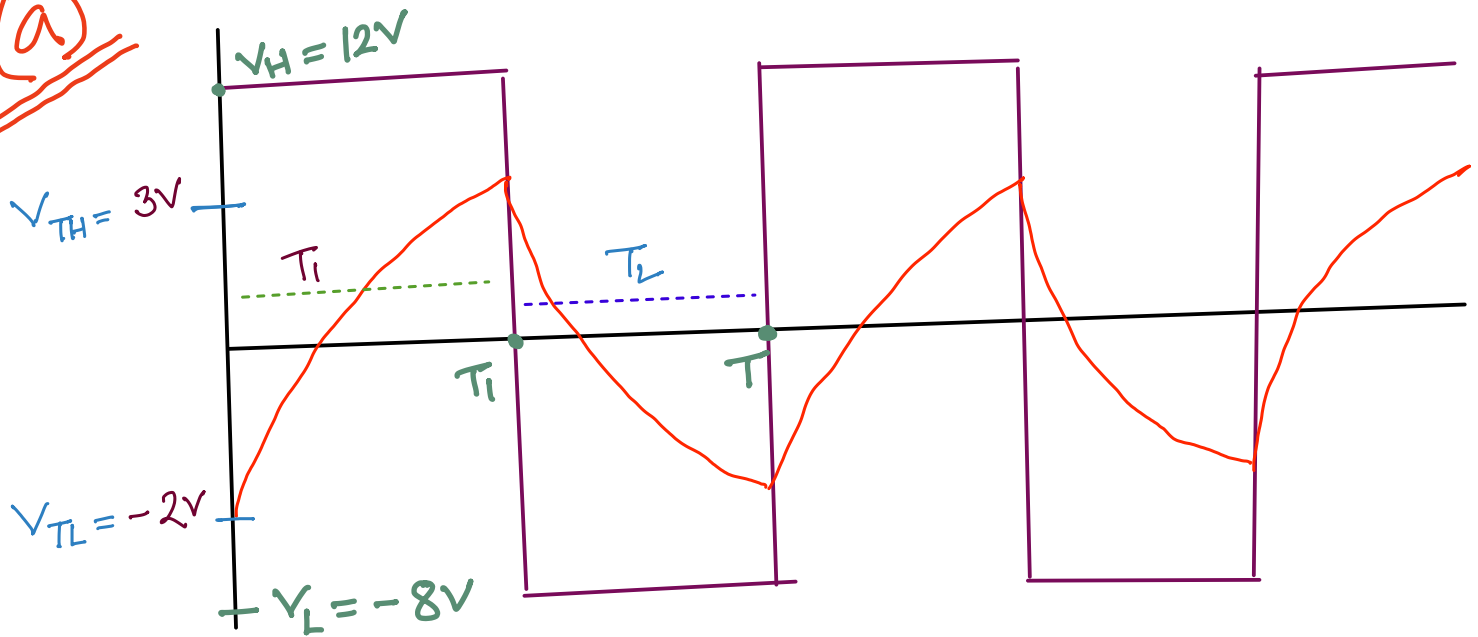
$$\therefore V_{Ref} = 3.846 \text{ volts}$$

The design



Question-3

(a)



given, $f = 800\text{Hz} \Rightarrow T = \frac{1}{f} = \frac{1}{800} = 1.25 \times 10^{-3}\text{sec}$
 $= 1.25\text{ms}$

Now, $T = T_H + T_L$

$$= T \ln \frac{V_H - V_{TL}}{V_H - V_{TH}} + T \ln \frac{V_L - V_{TH}}{V_L - V_{TL}}$$

$$\Rightarrow 1.25 \times 10^{-3} = T \left[\ln \frac{12+2}{12-3} + \ln \frac{-8-3}{-8+2} \right]$$

$$\Rightarrow 1.25 \times 10^{-3} = T \times 1.048$$

$$\therefore T = R_x \cdot C_x = 1.192 \times 10^{-3}\text{sec}$$

$$\begin{aligned}
 \text{Now, high time, } T_1 &= J \cdot \ln \frac{V_H - V_{TL}}{V_H - V_{TH}} \\
 &= 1.192 \times 10^{-3} \cdot \ln \frac{12 + 2}{12 - 3} \\
 &= 5.27 \times 10^{-4} \text{ sec} \\
 &= 0.527 \text{ ms}
 \end{aligned}$$

$$\begin{aligned}
 \text{Now, duty cycle, DC} &= \frac{T_1}{T} \times 100\% \\
 &= \frac{0.527}{1.25} \times 100\% \\
 &= \boxed{42.16\%}
 \end{aligned}$$

Now,

$$V_{TH} = \frac{R_1}{R_1 + R_2} V_H$$

$$\Rightarrow 3 = \frac{R_1}{R_1 + R_2} \cdot 12$$

$$\Rightarrow \frac{1}{4} = \frac{R_1}{R_1 + R_2}$$

$$\Rightarrow 4R_1 = R_1 + R_2$$

$$\Rightarrow 3R_1 = R_2$$

$$\therefore \frac{R_1}{R_2} = \frac{1}{3}$$

$$V_{TL} = \frac{R_1}{R_1 + R_2} V_L$$

$$\Rightarrow -2 = \frac{R_1}{R_1 + R_2} (-8)$$

$$\Rightarrow \frac{1}{4} = \frac{R_1}{R_1 + R_2}$$

$$\therefore \frac{R_1}{R_2} = \frac{1}{3}$$

(b) Now, given $f = 800 \times 2$
 $= 1600 \text{ Hz}$

V_H and V_L values are
 not given, Only
 $V_L = -V_H$ given

time period

$$\therefore T = \frac{1}{f} = \frac{1}{1600} = 6.25 \times 10^{-4} \text{ sec}$$

$$= 0.625 \text{ ms}$$

Now, $V_{TH} = -\frac{R_1}{R_2} V_L$

$$\Rightarrow V_{TH} = -\frac{2}{8} (-V_H)$$

$$\therefore V_{TH} = V_H/4$$

$$V_{TL} = -\frac{R_1}{R_2} V_H$$

$$\Rightarrow V_{TL} = -\frac{2}{8} V_H$$

$$\therefore V_{TL} = -V_H/4$$

Now,

$$T = T_1 + T_2$$

$$\Rightarrow T = T \left(\frac{V_{TH} - V_{TL}}{V_H} \right) + T \left(\frac{V_{TL} - V_{TH}}{V_L} \right)$$

$$\Rightarrow T = T \left(\frac{V_H/4 + V_H/4}{V_H} \right) + T \left(\frac{-V_H/4 - V_H/4}{-V_H} \right)$$

$$\Rightarrow 6.25 \times 10^{-4} \text{ sec} = 0.5 T + 0.5 T$$

$$\Rightarrow T = R_i \cdot C_i = 6.25 \times 10^{-4} \text{ sec}$$

$$\therefore C_i = \frac{6.25 \times 10^{-4} \text{ sec}}{20 \times 10^3 \Omega} = 3.125 \times 10^{-8} \text{ F} = 31.25 \text{ nF}$$

R_i value